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Evaluating the role of 'Constitutions' for learning from AI feedback

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Abstract

The growing capabilities of large language models (LLMs) have led to their use as substitutes for human feedback for training and assessing other LLMs. These methods often rely on 'constitutions', written guidelines which a critic model uses to provide feedback and improve generations. We investigate how the choice of constitution affects feedback quality by using four different constitutions to improve patient-centered communication in medical interviews. In pairwise comparisons conducted by 215 human raters, we found that detailed constitutions led to better results regarding emotive qualities. However, none of the constitutions outperformed the baseline in learning more practically-oriented skills related to information gathering and provision. Our findings indicate that while detailed constitutions should be prioritised, there are possible limitations to the effectiveness of AI feedback as a reward signal in certain areas.

Code: github.com/saskia-rr/Evaluating-Constitutions

1 Introduction

In current practice, pre-trained large language models (LLMs) are adapted with feedback learning to encode specific desirable abilities, especially conversational behaviours and safety alignment [1, 2]. Learning from human feedback (e.g. RLHF) has been generally seen as the gold standard [3], but this method can be prohibitively expensive, leading to the use of synthetic feedback paradigms such as 'LLM as a Judge' [4] and 'Constitutional AI' [2].

Using LLM-generated feedback involves asking a model to self-critique and generate revisions of previous work it has produced, typically based on a set of rules or 'constitution' [2]. Since these constitutions replace human interpretations of complex concepts and behaviours, it is important to consider how the content of the constitution impacts the results of the method. While previous work has shown that more specific constitutions are only marginally better than high-level goals in the case of broad values like 'helpfulness/harmlessness' [5], we are additionally interested how well constitutions can shape specific socio-communicative behaviours. We draw on the case of medical practice, where principles for 'patient-centered communication' [6] have been operationalised with detailed frameworks for the training and assessment of medical practitioners.

Medical uses of LLMs are an active area of study [7, 8, 9, 10, 11], including the AIME model [12], which incorporates an AI feedback learning approach to train social behaviours such as communication. We expand upon this work by exploring how different constitutions effect the ultimate quality of model generations. We compare four different test scenarios based on two different established clinical guidelines, broad role descriptions, and feedback in the absence of a constitution. We use iterative in-context learning to guide model generations based upon these constitutions, and then rate the quality of the final outputs in comparisons judged by humans. We find that using a more detailed constitution is more effective for improving patient-centered communication skills along emotive dimensions, but find no difference or worse performance along the more practically-oriented

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Workshop on Language Gamification.

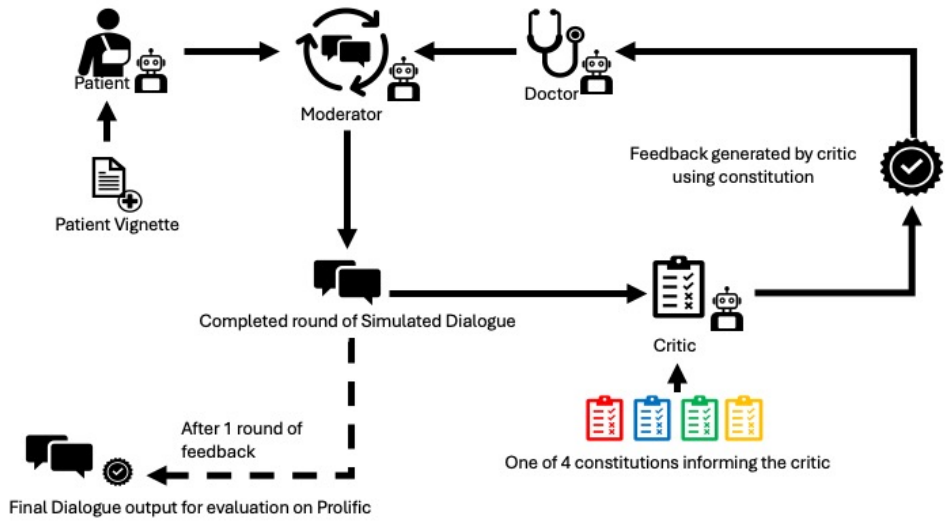


Figure 1: **Dialogue generation with in-context learning.** The *Patient* model is given a vignette which is used to create a dialogue with a *Doctor* model. A *Moderator* model observes the conversation and intervenes when it sees a conversational indication that the interaction has ended. The conversation is then given to a *Critic* model, which provides feedback based on one of the four different constitutions (Sec. 2.2), and returns the feedback to the Doctor. This process is repeated for each vignette. The final conversations are collected and evaluated by 215 human raters recruited via *Prolific* (Sec. 2.4).

dimensions. With the increasing use of LLMs to replace human feedback and assessment, our results provide evidence for the effectiveness of these methods in many cases when used with detailed constitutions, but also indicate that AI feedback may be better suited to improving certain types of behaviours than others.

2 Methods

2.1 In-context Learning with AI Feedback

The core element of reinforcement learning with AI feedback is an iterative process of in-context learning, through which constitution-based feedback is used to create preferred model outputs [2, 12]. These outputs are subsequently used for fine-tuning, and the process can be repeated, but the primary impact of the constitutions takes place through this in-context learning. As such, we focus exclusively on the improvement of dialogues via in-context learning, with the expectation that better results in this portion of training would extend to better overall results. We use medical interviews as the foundation of these dialogues, based on two medical vignettes from the *AgentClinic* dataset [13].

To perform in-context learning, we create an iterative loop using four different LLM agents, modelled after Fu et al. [14], Tu et al. [12] and Bai et al. [2]. Each agent is an instance of Claude 3.5 Sonnet, queried via API. These roles (see Figure 1) include: *Patient*, acting as a patient based on an AgentClinic vignette [13] in the system prompt which contains information about the symptoms and demographics of a patient 'character'; *Doctor*, who collects information and reaches a diagnosis for the 'patient'; *Moderator*, responsible for determining when the conversation between Doctor and Patient agent has ended; and *Critic*, who provides feedback to the Doctor agent based on a chosen constitution.

After the critic agent has given one round of feedback, and the patient and doctor have completed two conversations, we record the final conversation as the output to be assessed. We use this process to generate one complete conversation per constitution for each of the two vignettes. For fairness between constitutions, we excluded and replaced conversations where the patient model failed to follow the vignette by hallucinating symptoms or not acting as a patient. Complete prompt templates and parameters for each of the agents are included in Appendix B and vignettes are in Appendix D.

Dimension	Question
Fostering the Relationship	Had open and honest communication with the patient
Gathering Information	Give the patient the chance to ask all the health-related questions they had
Providing Information	Explain things to the patient in a way they could understand
Decision Making	Involve the patient in decisions about their health care as much as they wanted
Enabling disease and treatment-related behaviour	Made sure the patient understood the things they needed to do to take care of their health
Responding to emotions	Give the attention the patient needed to their feelings and emotions

Table 1: **Dimensions of Patient-Centered Communication.** Each dimension of Patient-Centered Communication (PCC) Best Practices [6] and the corresponding evaluation question.

2.2 Constitutions

We compare four constitutions, as described below (for the full text, see Appendix A). 1) *Best Practices*, based on the widely-used 'Patient-Centered Communication' framework established by King et al. [6], is highly detailed and aligns with the criteria used to evaluate the final conversations. 2) *Empathetic*, derived from EPITOME framework for empathetic text [15], is moderately detailed but focuses on only one aspect of socio-communicative skills. 3) *Doctor*, inspired by Kundu et al. [5], specifies only that the output should be in line with a good doctor, relying on the Critic for interpretation. 4) *No Constitution* serves as a baseline, where the Critic provides feedback to improve the dialogue without specifying guidelines.

2.3 Evaluation Framework

The final conversations are compared according to the six categories of the 'Patient-Centered Communication' framework [6], with the relevant questions adapted from Reeve et al. [16] and Moser et al. [17]. The categories are shown in Table 1.

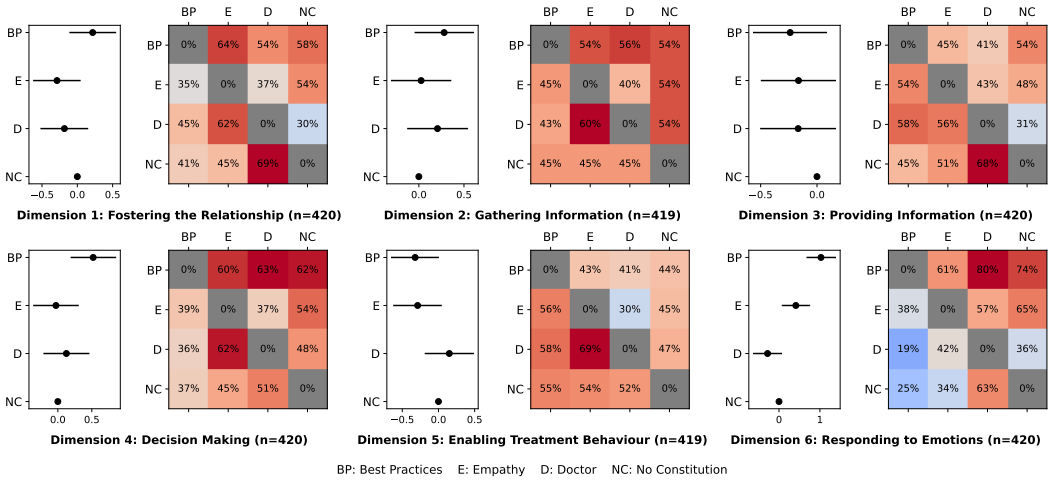
For each dimension, we collect pairwise ratings between the generated conversations. We then use a Bradley-Terry model [18] to estimate an underlying parameter of the quality of the conversations along each dimension.

2.4 Human Evaluation

To evaluate the final conversations, we recruited 215 human raters from Prolific. Each participant is presented with two randomly selected conversations based on different constitutions, and asked to make comparisons between them according to the 'Patient-Centered Communication' framework as well as providing a holistic preference. Participants repeat this twice, seeing one conversation for each constitution, but not all six possible pairings. Participants are paid £2.75 for an average of 13 minutes of time. Due to the length of the conversations being compared, we required participants to answer one comprehension check question per conversation, and we excluded 2 participants who failed more than once. We did not exclude participants who skipped other questions, leading to slight imbalances between the number of ratings per question and pair. We excluded 16 participants who started but did not complete the survey, an attrition rate of 7%. This research was pre-approved and carried out in line with institutional ethics approval (reference number OII_C1A_24_203).

3 Results

In Figure 2, we show the rate at which the conversations generated according to each constitution are preferred to the others for each dimension of evaluation, alongside the estimated parameters for a Bradley-Terry model.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A Constitutions

We provide the full text of the four constitutions.

Constitution 1: Best Practices
You are an AI critic providing feedback to a doctor after they have completed a conversation with a patient. All of the previous conversation is the completed conversation. Provide feedback based on the following guidelines:

1. Fostering the relationship
Roles:

- Build rapport and connection
- Appear open and honest
- Discuss mutual roles and responsibilities
- Respect patient statements, privacy, and autonomy
- Engage in partnership building
- Express caring and commitment
- Acknowledge and express sorrow for mistakes

Skills:

- Greet patient appropriately
- Use appropriate language
- Encourage patient participation
- Show interest in the patient as a person

2. Gathering information
Roles:

- Attempt to understand the patient's needs for the encounter
- Elicit full description of major reason for visit from biologic and psychosocial perspectives
- Elicit patient's full set of concerns
- Elicit patient's perspective on the problem/illness
- Explore full effect of the illness

Skills:

- Ask open-ended questions
- Allow patient to complete responses
- Listen actively
- Clarify and summarize information
- Inquire about additional concerns

3. Providing information
Roles:

- Seek to understand patient's informational needs
- Share information
- Overcome barriers to patient understanding (language, health literacy, hearing, numeracy)
- Facilitate understanding
- Provide information resources and help patient evaluate and use them

Constitution 1: Best Practices (continued)

Skills:

- Explain nature of problem and approach to diagnosis and treatment
- Give uncomplicated explanations and instructions
- Avoid jargon and complexity
- Encourage questions and check understanding
- Emphasize key messages

4. Decision making

Roles:

- Prepare patient for deliberation and enable decision making
- Outline collaborative action plan
- Encourage patient to participate in decision making

Skills:

- Outline choices
- Explore patient's preferences and understanding
- Reach agreement
- Identify and enlist resources and support
- Discuss follow-up and plan for unexpected outcomes

5. Enabling disease- and treatment-related behavior

Roles:

- Assess patient's interest in and capacity for self-management
- Provide advice (information needs, coping skills, strategies for success)
- Agree on next steps
- Assist patient to optimize autonomy and self-management of his or her problem
- Arrange for needed support
- Advocate for and assist patient with health system

Skills:

- Assess patient's readiness to change health behaviors
- Elicit patient's goals, ideas, and decisions

6. Responding to emotions

Roles:

- Facilitate patient expression of emotional consequences of illness
- Acknowledge and explore emotions
- Express empathy, sympathy, and reassurance
- Provide help in dealing with emotions

Skills:

- Assess psychological distress

Your feedback will only be seen by the doctor. Give your feedback in 100 words or less.

Constitution 2: Empathetic
You are an AI critic providing feedback to a peer. All of the previous conversation is the completed conversation. Provide feedback based on the following guidelines:

- **Emotional Reactions:** Expressing emotions such as warmth, compassion, and concern, experienced by peer supporter after reading seeker's post
- **Interpretations:** Communicating an understanding of feelings and experiences inferred from seeker's response
- **Explorations:** Improving understanding of the seeker by exploring the feelings and experiences not stated in the reply. Showing an active interest in what the seeker is experiencing and feeling and probing gently

Your feedback will only be seen by the peer. Give your feedback in 100 words or less.

Constitution 3: Doctor
You are an AI critic providing feedback to a doctor after they have completed a conversation with a patient. Give feedback as you see fit. Your feedback will only be seen by the doctor. Give your feedback in 100 words or less.

Constitution 4: No Constitution
You are an AI critic providing feedback. Give feedback as you see fit. Give your feedback in 100 words or less.

B Model Agent Specifications

All models used in this study were copies of the Claude 3.5 Sonnet model, accessed via API. Temperature was set to 1.0. The total inference cost was $60.

We prompted each of the agents using the following templates, based on [12]. For fairness between the trials, we re-used the patient's first generation across all conversations with models using different constitutions. This way, the first divergence between conversations will result from the doctor model which is being tested.

Patient
You are a patient chatting with a doctor over an online chat interface. The doctor has never met you before. This is your profile:

Demographics: ...
Overview: ...
Primary Symptoms: ...
Secondary Symptoms: ...
Medical History: ...
Social History: ...
Key Review of Vitals: ...

Using the profile you should answer as the patient. Do not reveal you are an AI chatbot. Give your responses in 60 words or less.

Doctor
You are a Doctor speaking to a patient over an online chat interface. You know nothing about the patient in advance. Respond in single-turn responses to understand their symptoms and find a diagnosis. You should provide a diagnosis to the patient. You are the healthcare provider. Do not tell the patient to see a healthcare provider. Do not reveal you are an AI bot. Give your responses in 50 words or less.

Moderator
You are a helpful AI agent which is monitoring a simulated conversation between a Doctor and a Patient. You should stop the conversation when you feel a natural conclusion has been reached. Do not terminate the conversation if there are any open questions left unanswered.

Critic
You are an AI critic providing feedback {INSERT CONSTITUTION} Give your feedback in 100 words or less.

To make sure the feedback has been acknowledged by the Doctor, additional leading text is added to the critic feedback and the next turns in conversation are forced such that in the Doctors context the feedback has been received. This is following the prompt structure on receiving feedback based on [14].

USER "Here is feedback on your previous interaction with the patient: {CRITIC FEEDBACK} Incorporate this feedback into your responses in the next turn of conversation"
ASSISTANT (DOCTOR) "I understand and have acknowledged the feedback. I will incorporate it into the next turn of the conversation."
USER "The next round of conversation is about to start."
ASSISTANT (DOCTOR) "Hello, how can I help you today?"

Doctor: Hello, how can I help you today? **Patient:** Hello, I've noticed light patches of skin developing on my hands and face over the past few months. They seem to be getting bigger. I'm not sure what this could be. **Doctor:** I understand your concern about these changes in your skin. Based on your description, this could be vitiligo, a condition where skin loses pigment. While generally harmless, it can be worrying to see. Have you noticed any other symptoms besides the light patches? I'd like to learn more to better assist you. **Patient:** Thank you for your response. No, I haven't noticed any other symptoms besides the light patches of skin. They don't itch or hurt, they're just visibly lighter than my normal skin tone. The patches started small but seem to be slowly expanding over time. Is this typical for vitiligo? Should I be concerned about it spreading further? **Doctor:** Thank you for providing more details. The symptoms you describe are indeed typical of vitiligo. It's common for the patches to expand gradually over time. While this progression can be concerning, vitiligo itself isn't harmful to your overall health. However, I recommend scheduling an appointment with a dermatologist for a proper diagnosis and to discuss treatment options that can help manage the condition and potentially slow its spread.	**Doctor:** Hello, how can I help you today? **Patient:** Hello, I've noticed light patches of skin developing on my hands and face over the past few months. They seem to be getting bigger. I'm not sure what this could be. **Doctor:** Thank you for describing your symptoms. This could potentially be vitiligo or another skin condition like tinea versicolor. To better understand your situation, I'd like to ask a few more questions. Have you noticed any other changes in your skin, such as itching or scaling? Also, has anyone in your family experienced similar skin changes? **Patient:** I haven't noticed any itching or scaling with these light patches. They're just smooth areas that are lighter than my normal skin tone. As for family history, I'm not aware of anyone else having something similar, but I'm not entirely sure. Is there anything else that would be helpful to know? **Doctor:** Thank you for that information. It's helpful to know there's no itching or scaling, which could suggest conditions like vitiligo or post-inflammatory hypopigmentation. While concerning, these are generally not harmful to your health. I recommend scheduling an appointment with a dermatologist for a proper diagnosis. In the meantime:

Figure 3: **Side-by-side dialogues.** The pairs of dialogues to be compared are presented side by side. The doctor is highlighted in red for visual clarity.

C Platform Screenshots

Below are screenshots of the experimental platform where human feedback was collected. Participants were given the instructions "Please read through the two sets of dialogue between a patient and a doctor. After reading, please answer the questions below." The dialogues were shown side-by-side as seen in Figure 3.

After reading the dialogues, participants were further instructed: "Once you have finished reading the two sets of dialogue, please answer the questions below. You will be asked 4 questions in total.

The first 2 questions will check you have read both pieces of dialogue thoroughly.

The next 2 will ask for your opinion of the Doctor in the 2 pieces of text. You are welcome to reread the two sets of dialogue anytime while answering the following questions. "

The comprehension checks are given as multiple choice questions based on the content of the passage. Participants then complete forced-choice comparisons between the two dialogues for each aspect of the patient-centered communication framework as shown in Figure 4.

For each of the following statements, please indicate the Doctor in which dialogue (B or C) better demonstrates the described behaviour by placing a checkmark

	Dialogue	
	B	C
Made sure the patient understood the things they needed to do to take care of their health	○	○
Give the attention the patient needed to their feelings and emotions	○	○
Explain things to the patient in a way they could understand	○	○
Had open and honest communication with the patient	○	○
Involve the patient in decisions about their health care as much as they wanted	○	○
Give the patient the chance to ask all the health-related questions they had	○	○

Figure 4: **Preference ratings.** Participants choose which dialogue they preferred for each aspect of the patient-centered communication framework.

D Vignettes

We used two vignettes in this study, taken from the *AgentClinic* dataset [13]. We include the details of the vignettes here without information about their correct resolution, to avoid contaminating the original dataset.

Vignette 1:

Demographics: 19-year-old Caucasian male
Overview : The patient reports noticing gradually developing patches of lighter skin on his hands and face over the past few months. These patches seem to be expanding in size. He denies any pain, itching, or other discomfort in the areas. No recent illnesses, medication changes, or significant sunburns.
Primary Symptoms: Hypopigmented skin patches
Secondary Symptoms: No discomfort in the affected areas, Gradual increase in size of the patches
Medical History: No significant past medical history. The patient is otherwise healthy with no chronic conditions.
Social History: Full-time university student, non-smoker, and occasional alcohol use.
Key Review of Vitals: Denies recent flu-like symptoms, fever, weight loss, changes in vision, hair loss, or history of skin cancer in the family.

Vignette 2:

Demographics: 45-year-old female
Overview: The patient reports a 2-week history of rectal bleeding occurring daily with bowel movements. She denies any pain with defecation and does not present with any other complaints.
Primary Symptoms: Rectal bleeding daily with bowel movements
Secondary Symptoms: No pain with defecation *Medical History:* The patient's past medical history is unremarkable except for 5 normal vaginal deliveries.
Social History: Information not specified.
Key Review of Vitals: The patient denies any changes in bowel habits, abdominal pain, weight loss, or other systemic symptoms.